BMI-500

Hypothesis - HW1 : SIR & SEIR model

Goal: Build SIR & SEIR model with numerical solver to see how an outbreak grows, peaks & ends.

(i) When and how big is infection peak : Peak occurs when new infection per day matches recovery per day. Sickness stays for 10 days ($1/\gamma$) and infects $\beta N / \beta S \approx 0.25$ people per day. $R_0 = \beta N/\gamma = 2.5$

So 2-3 people are infected by each infected person & outbreak grow

Net change $\approx 0.25 - 0.1 = 0.15$ per sick person per day

So sick group should grow around 15% per day so roughly double every 5 days. Peak occurs at $\beta \times S = 0.1 \Rightarrow S = 800$

So when 1200 are infected & 800 are susceptible. Going by doubling rate, it occurs around the 35-40 days timeframe

(ii) ∴ expect several waves. Births keep adding new healthy people so after first wave the disease can spread again. Over time it should die down to steady levels.

(iii) ∴ expect β to affect the peak more since it controls how fast the disease spreads. Increasing it should make the peak higher and arrive earlier.